GVPT Data Science Camp

Data Manipulation

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Learning Objectives

- Understanding basic operations in `R`

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- Understanding basic operations in `R`
- Introduction to `dplyr`

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- Understanding basic operations in `R`
- Introduction to `dplyr`
- Cleaning and manipulating data

Objects and Functions

Objects

- Create new objects with <-

```
x <- 6 * 8

x
```

```
x <- 2 * 15

x
```

Functions

- Several functions come pre-installed:

```r
seq(15, 30)
```

- Several functions come pre-installed:

```r
seq(15, 30)
```

- You can use functions to create objects:

```r
x <- seq(15, 30)

x
```

Introduction to Data Manipulation

What is Data Manipulation?

- Data rarely arrive in exactly the form you need for analysis

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- Data manipulation is reshaping data to answer your question: picking out rows and columns, creating new variables, and computing summaries

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- Data rarely arrive in exactly the form you need for analysis
- Data manipulation is reshaping data to answer your question: picking out rows and columns, creating new variables, and computing summaries
- Today's tool: the `dplyr` package from the tidyverse

Exercise

- Load the packages we used yesterday:

```r
library(tidyverse)
library(readxl)
```

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- Load the packages we used yesterday:

```
library(tidyverse)
library(readxl)
```

- Re-run yesterday's setup code (`read_excel()`, `rename()`, `select()`, and `mutate()`) to re-create the `cel` object, then view the data:

```
view(cel)
```

Data Types

- In `cel`:

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- `fct` stands for factors, which `R` uses to represent categorical variables with fixed values — like `benchmark` (Below, Meets, Exceeds)

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- `dbl` stands for doubles — like `les` and `ideology`

Data Types

- In `cel`:
- `fct` stands for factors, which `R` uses to represent categorical variables with fixed values — like `benchmark` (Below, Meets, Exceeds)
- `dbl` stands for doubles — like `les` and `ideology`
- `chr` stands for strings (character vectors) — like `member`, `state`, and `party`

Data Types

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- `dttm` stands for date-times (a date + time)

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- `int` stands for integer
- `dttm` stands for date-times (a date + time)
- `lgl` stands for logical, vectors that contain only `TRUE` or `FALSE`

- Five basic functions:

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- `filter()`

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- `arrange()`

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- `select()`

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- `filter()`
- `arrange()`
- `select()`
- `mutate()`

Introduction to dplyr

- Five basic functions:
- `filter()`
- `arrange()`
- `select()`
- `mutate()`
- `summarise()`

Filtering and Arranging Data

```r
filter(cel, state == "MD", congress > 110)
```

```
   member              congress  year state   dem female votepct ideology seniority  born   les bills_passed benchmark party
   <chr>                  <dbl> <dbl> <chr> <dbl>  <dbl>   <dbl>    <dbl>     <dbl> <dbl> <dbl>        <dbl> <fct>     <chr>
 1 Bartlett, Roscoe         111  2009 MD        0      0      58   0.493          9  1926 0.172            0 Below     Republican
 2 Cummings, Elijah         111  2009 MD        1      0      80  -0.438          7  1951 1.74             5 Meets     Democrat
 3 Edwards, Donna           111  2009 MD        1      1      86  -0.562          1  1958 0.101            0 Below     Democrat
 4 Hoyer, Steny             111  2009 MD        1      0      74  -0.380         15  1939 1.04             3 Meets     Democrat
 5 Kratovil, Frank          111  2009 MD        1      0      49  -0.0370         1  1968 1.56             2 Exceeds   Democrat
 6 Ruppersberger, C.        111  2009 MD        1      0      72  -0.295          4  1946 0.720            1 Meets     Democrat
 7 Sarbanes, John           111  2009 MD        1      0      70  -0.468          2  1962 1.95             4 Exceeds   Democrat
 8 Van Hollen, Chris        111  2009 MD        1      0      75  -0.391          4  1959 1.61             2 Exceeds   Democrat
 9 Bartlett, Roscoe         112  2011 MD        0      0      61   0.493         10  1926 0.107            0 Below     Republican
10 Cummings, Elijah         112  2011 MD        1      0      75  -0.438          8  1951 0.443            0 Meets     Democrat
# ℹ 54 more rows
# ℹ Use `print(n = ...)` to see more rows
```

- First argument must always be a data frame

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- The arguments that follow describe which columns to operate on, using the variable names (without quotes)

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- The arguments that follow describe which columns to operate on, using the variable names (without quotes)
- Output is always a new data frame

```r
filter(cel, state %in% c("MD", "VA"))
```

```
   member            congress  year state   dem female votepct ideology seniority  born    les bills_passed benchmark party
   <chr>                <dbl> <dbl> <chr> <dbl>  <dbl>   <dbl>    <dbl>     <dbl> <dbl>  <dbl>        <dbl> <fct>     <chr>
 1 Bauman, Robert          93  1973 MD        0      0      NA   0.533          1  1937 0.0620            0 Below     Republican
 2 Broyhill, Joel          93  1973 VA        0      0      56   0.159         11  1919 2.42              6 Exceeds   Republican
 3 Butler, M.              93  1973 VA        0      0      55   0.332          1  1925 0.0114            0 Below     Republican
 4 Byron, Goodloe          93  1973 MD        1      0      65  -0.0140         2  1929 0.124             0 Below     Democrat
 5 Daniel, Robert          93  1973 VA        0      0      47   0.318          1  1936 0.0227            0 Below     Republican
 6 Daniel, W.              93  1973 VA        1      0     100   0.161          3  1914 0.0382            0 Below     Democrat
 7 Downing, Thomas         93  1973 VA        1      0      78  -0.0650         8  1919 1.94              4 Meets     Democrat
 8 Gude, Gilbert           93  1973 MD        0      0      64  -0.0280         4  1923 1.46              3 Exceeds   Republican
 9 Hogan, Lawrence         93  1973 MD        0      0      63   0.158          3  1928 0.200             0 Meets     Republican
10 Holt, Marjorie          93  1973 MD        0      1      59   0.368          1  1920 0.222             1 Meets     Republican
# ℹ 482 more rows
# ℹ Use `print(n = ...)` to see more rows
```

```r
filter(cel, votepct > 50 & votepct < 60)
```

```
   member              congress  year state   dem female votepct ideology seniority  born    les bills_passed benchmark party
   <chr>                  <dbl> <dbl> <chr> <dbl>  <dbl>   <dbl>    <dbl>     <dbl> <dbl>  <dbl>        <dbl> <fct>     <chr>
 1 Abdnor, James             93  1973 SD        0      0      55    0.241         1  1923 0.110             0 Below     Republican
 2 Abzug, Bella              93  1973 NY        1      1      56   -0.597         2  1920 0.762             1 Meets     Democrat
 3 Annunzio, Frank           93  1973 IL        1      0      53   -0.398         5  1915 0.628             2 Below     Democrat
 4 Arends, Leslie            93  1973 IL        0      0      57    0.301        20  1895 0.0155            0 Below     Republican
 5 Ashbrook, John            93  1973 OH        0      0      57    0.612         7  1928 0.172             0 Below     Republican
 6 Baker, LaMar              93  1973 TN        0      0      55    0.308         2  1915 0.819             1 Exceeds   Republican
 7 Beard, Robin              93  1973 TN        0      0      55    0.297         1  1939 0.0475            0 Below     Republican
 8 Bergland, Bob             93  1973 MN        1      0      59   -0.380         2  1928 0.125             0 Below     Democrat
 9 Brademas, John            93  1973 IN        1      0      55   -0.462         8  1927 5.62              7 Exceeds   Democrat
10 Breckinridge, John        93  1973 KY        1      0      52   -0.223         1  1913 0.484             1 Meets     Democrat
# ℹ 3,013 more rows
# ℹ Use `print(n = ...)` to see more rows
```

```r
filter(cel, votepct > 50 | votepct < 60)
```

```
   member           congress  year state   dem female votepct ideology seniority  born      les bills_passed benchmark party
   <chr>               <dbl> <dbl> <chr> <dbl>  <dbl>   <dbl>    <dbl>     <dbl> <dbl>    <dbl>        <dbl> <fct>     <chr>
 1 Abdnor, James          93  1973 SD        0      0      55   0.241          1  1923 0.110               0 Below     Republican
 2 Abzug, Bella           93  1973 NY        1      1      56  -0.597          2  1920 0.762               1 Meets     Democrat
 3 Adams, Brock           93  1973 WA        1      0      85  -0.396          5  1927 1.24                2 Meets     Democrat
 4 Addabbo, Joseph        93  1973 NY        1      0      75  -0.402          7  1925 0.155               0 Below     Democrat
 5 Albert, Carl           93  1973 OK        1      0      93  -0.392         14  1908 0.00103             0 Below     Democrat
 6 Alexander, Bill        93  1973 AR        1      0     100  -0.410          3  1934 1.88                2 Meets     Democrat
 7 Anderson, John         93  1973 IL        0      0      72   0.183          7  1922 0.289               0 Meets     Republican
 8 Anderson, Glenn        93  1973 CA        1      0      75  -0.269          3  1913 0.447               0 Meets     Democrat
 9 Andrews, Ike           93  1973 NC        1      0      50  -0.178          1  1925 0.0879              0 Below     Democrat
10 Andrews, Mark          93  1973 ND        0      0      73   0.0870         6  1926 0.106               0 Below     Republican
# ℹ 11,236 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Exercise

- Find all member-Congress observations where the member ran unopposed (received 100% of the vote)

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- Find all members from Maryland

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- Find all member-Congress observations where the member ran unopposed (received 100% of the vote)
- Find all members from Maryland
- Find all members from Maryland and Virginia

Exercise

- Find all member-Congress observations where the member ran unopposed (received 100% of the vote)
- Find all members from Maryland
- Find all members from Maryland and Virginia
- Find all member-Congress observations with a vote share greater than 60% and less than 70%

Exercise

- Find all member-Congress observations where the member ran unopposed (received 100% of the vote)
- Find all members from Maryland
- Find all members from Maryland and Virginia
- Find all member-Congress observations with a vote share greater than 60% and less than 70%
- Find all member-Congress observations with a vote share less than 60% or greater than 70%

Check your Answers

```r
filter(cel, votepct == 100)
```

```
   member           congress  year state   dem female votepct ideology seniority  born   les bills_passed benchmark party
   <chr>               <dbl> <dbl> <chr> <dbl>  <dbl>   <dbl>    <dbl>     <dbl> <dbl> <dbl>        <dbl> <fct>     <chr>
 1 Alexander, Bill        93  1973 AR        1      0     100 -0.410           3  1934 1.88             2 Meets     Democrat
 2 Boland, Edward         93  1973 MA        1      0     100 -0.306          11  1911 2.05             2 Meets     Democrat
 3 Breaux, John           93  1973 LA        1      0     100 -0.123           1  1944 1.16             1 Exceeds   Democrat
 4 Brinkley, Jack         93  1973 GA        1      0     100 -0.0730          4  1930 0.217            0 Below     Democrat
 5 Burke, James           93  1973 MA        1      0     100 -0.400           8  1910 0.789            0 Meets     Democrat
 6 Burleson, Omar         93  1973 TX        1      0     100  0.00200        14  1906 0.130            0 Below     Democrat
 7 Conte, Silvio          93  1973 MA        0      0     100 -0.0230          8  1921 1.46             3 Exceeds   Republican
 8 Daniel, W.             93  1973 VA        1      0     100  0.161           3  1914 0.0382           0 Below     Democrat
 9 Donohue, Harold        93  1973 MA        1      0     100 -0.296          14  1901 1.42             3 Meets     Democrat
10 Duncan, John           93  1973 TN        0      0     100  0.194           5  1919 0.257            0 Meets     Republican
# ℹ 740 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Check your Answers

```r
filter(cel, state == "MD")
```

```
   member             congress  year state   dem female votepct ideology seniority  born    les bills_passed benchmark party
   <chr>                 <dbl> <dbl> <chr> <dbl>  <dbl>   <dbl>    <dbl>     <dbl> <dbl>  <dbl>        <dbl> <fct>     <chr>
 1 Bauman, Robert           93  1973 MD        0      0      NA   0.533          1  1937 0.0620            0 Below     Republican
 2 Byron, Goodloe           93  1973 MD        1      0      65  -0.0140         2  1929 0.124             0 Below     Democrat
 3 Gude, Gilbert            93  1973 MD        0      0      64  -0.0280         4  1923 1.46              3 Exceeds   Republican
 4 Hogan, Lawrence          93  1973 MD        0      0      63   0.158          3  1928 0.200             0 Meets     Republican
 5 Holt, Marjorie           93  1973 MD        0      1      59   0.368          1  1920 0.222             1 Meets     Republican
 6 Long, Clarence           93  1973 MD        1      0      66  -0.274          6  1908 0.210             0 Below     Democrat
 7 Mills, William           93  1973 MD        0      0      NA   0.210          2  1924 0.0517            0 Below     Republican
 8 Mitchell, Parren         93  1973 MD        1      0      80  -0.526          2  1922 0.0920            0 Below     Democrat
 9 Sarbanes, Paul           93  1973 MD        1      0      70  -0.435          2  1933 0.0103            0 Below     Democrat
10 Bauman, Robert           94  1975 MD        0      0      53   0.533          2  1937 0.0803            0 Below     Republican
# ℹ 201 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Check your Answers

```r
filter(cel, state %in% c("MD", "VA"))
```

```
   member            congress  year state   dem female votepct ideology seniority  born    les bills_passed benchmark party
   <chr>                <dbl> <dbl> <chr> <dbl>  <dbl>   <dbl>    <dbl>     <dbl> <dbl>  <dbl>        <dbl> <fct>     <chr>
 1 Bauman, Robert          93  1973 MD        0      0      NA   0.533          1  1937 0.0620            0 Below     Republican
 2 Broyhill, Joel          93  1973 VA        0      0      56   0.159         11  1919 2.42              6 Exceeds   Republican
 3 Butler, M.              93  1973 VA        0      0      55   0.332          1  1925 0.0114            0 Below     Republican
 4 Byron, Goodloe          93  1973 MD        1      0      65  -0.0140         2  1929 0.124             0 Below     Democrat
 5 Daniel, Robert          93  1973 VA        0      0      47   0.318          1  1936 0.0227            0 Below     Republican
 6 Daniel, W.              93  1973 VA        1      0     100   0.161          3  1914 0.0382            0 Below     Democrat
 7 Downing, Thomas         93  1973 VA        1      0      78  -0.0650         8  1919 1.94              4 Meets     Democrat
 8 Gude, Gilbert           93  1973 MD        0      0      64  -0.0280         4  1923 1.46              3 Exceeds   Republican
 9 Hogan, Lawrence         93  1973 MD        0      0      63   0.158          3  1928 0.200             0 Meets     Republican
10 Holt, Marjorie          93  1973 MD        0      1      59   0.368          1  1920 0.222             1 Meets     Republican
# ℹ 482 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Check your Answers

```r
filter(cel, votepct > 60 & votepct < 70)
```

```
   member             congress  year state   dem female votepct ideology seniority  born    les bills_passed benchmark party
   <chr>                 <dbl> <dbl> <chr> <dbl>  <dbl>   <dbl>    <dbl>     <dbl> <dbl>  <dbl>        <dbl> <fct>     <chr>
 1 Armstrong, William       93  1973 CO        0      0      62   0.508          1  1937 0.137             1 Meets     Republican
 2 Ashley, Thomas           93  1973 OH        1      0      69  -0.350         10  1923 3.76              5 Exceeds   Democrat
 3 Aspin, Les               93  1973 WI        1      0      64  -0.320          2  1938 0.472             0 Meets     Democrat
 4 Bafalis, L.              93  1973 FL        0      0      62   0.303          1  1929 0.0517            0 Below     Republican
 5 Barrett, William         93  1973 PA        1      0      66  -0.469         13  1896 1.65              2 Meets     Democrat
 6 Bell, Alphonzo           93  1973 CA        0      0      61   0.183          7  1914 0.233             0 Meets     Republican
 7 Biester, Edward          93  1973 PA        0      0      64   0.0350         4  1931 0.115             0 Below     Republican
 8 Bolling, Richard         93  1973 MO        1      0      63  -0.504         13  1916 0                 0 Below     Democrat
 9 Bowen, David             93  1973 MS        1      0      62  -0.134          1  1932 0.588             1 Meets     Democrat
10 Brasco, Frank            93  1973 NY        1      0      64  -0.423          4  1932 0.585             1 Meets     Democrat
# ℹ 3,413 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Check your Answers

```r
filter(cel, votepct < 60 | votepct > 70)
```

```
   member            congress  year state   dem female votepct ideology seniority  born     les bills_passed benchmark party
   <chr>                <dbl> <dbl> <chr> <dbl>  <dbl>   <dbl>    <dbl>     <dbl> <dbl>   <dbl>        <dbl> <fct>     <chr>
 1 Abdnor, James           93  1973 SD        0      0      55   0.241          1  1923 0.110              0 Below     Republican
 2 Abzug, Bella            93  1973 NY        1      1      56  -0.597          2  1920 0.762              1 Meets     Democrat
 3 Adams, Brock            93  1973 WA        1      0      85  -0.396          5  1927 1.24               2 Meets     Democrat
 4 Addabbo, Joseph         93  1973 NY        1      0      75  -0.402          7  1925 0.155              0 Below     Democrat
 5 Albert, Carl            93  1973 OK        1      0      93  -0.392         14  1908 0.00103            0 Below     Democrat
 6 Alexander, Bill         93  1973 AR        1      0     100  -0.410          3  1934 1.88               2 Meets     Democrat
 7 Anderson, John          93  1973 IL        0      0      72   0.183          7  1922 0.289              0 Meets     Republican
 8 Anderson, Glenn         93  1973 CA        1      0      75  -0.269          3  1913 0.447              0 Meets     Democrat
 9 Andrews, Ike            93  1973 NC        1      0      50  -0.178          1  1925 0.0879             0 Below     Democrat
10 Andrews, Mark           93  1973 ND        0      0      73   0.0870         6  1926 0.106              0 Below     Republican
# ℹ 7,132 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Important Features

- `==` is equal to

Important Features

- `==` is equal to
- `!=` is not equal to

Important Features

- `==` is equal to
- `!=` is not equal to
- `>=` is greater than or equal to

Important Features

- `==` is equal to
- `!=` is not equal to
- `>=` is greater than or equal to
- `<=` is less than or equal to

Important Features

- $|$ is or

Important Features

- `|` is or
- `&` is AND

Important Features

- `|` is or
- `&` is AND
- `%in%` is in

Arrange Rows with arrange()

```r
arrange(cel, member, congress)
```

```
   member              congress  year state   dem female votepct ideology seniority  born    les bills_passed benchmark party
   <chr>                  <dbl> <dbl> <chr> <dbl>  <dbl>   <dbl>    <dbl>     <dbl> <dbl>  <dbl>        <dbl> <fct>     <chr>
 1 Abdnor, James             93  1973 SD        0      0      55    0.241         1  1923 0.110             0 Below     Republican
 2 Abdnor, James             94  1975 SD        0      0      68    0.241         2  1923 0.186             0 Meets     Republican
 3 Abdnor, James             95  1977 SD        0      0      70    0.241         3  1923 0.428             1 Meets     Republican
 4 Abdnor, James             96  1979 SD        0      0      NA    0.241         4  1923 0.313             1 Meets     Republican
 5 Abercrombie, Neil         99  1985 HI        1      0      NA   -0.432         1  1938 0.0154            0 Below     Democrat
 6 Abercrombie, Neil        102  1991 HI        1      0      60   -0.432         1  1938 1.10              0 Exceeds   Democrat
 7 Abercrombie, Neil        103  1993 HI        1      0      73   -0.432         2  1938 0.0982            0 Below     Democrat
 8 Abercrombie, Neil        104  1995 HI        1      0      54   -0.432         3  1938 0.693             1 Exceeds   Democrat
 9 Abercrombie, Neil        105  1997 HI        1      0      50   -0.432         4  1938 0.235             0 Meets     Democrat
10 Abercrombie, Neil        106  1999 HI        1      0      62   -0.432         5  1938 0.427             1 Meets     Democrat
# ℹ 11,596 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Arrange Rows with arrange()

```r
arrange(cel, member, desc(congress))
```

```
   member              congress  year state   dem female votepct ideology seniority  born   les bills_passed benchmark party
   <chr>                  <dbl> <dbl> <chr> <dbl>  <dbl>   <dbl>    <dbl>     <dbl> <dbl> <dbl>        <dbl> <fct>     <chr>
 1 Abdnor, James             96  1979 SD        0      0      NA    0.241         4  1923 0.313            1 Meets     Republican
 2 Abdnor, James             95  1977 SD        0      0      70    0.241         3  1923 0.428            1 Meets     Republican
 3 Abdnor, James             94  1975 SD        0      0      68    0.241         2  1923 0.186            0 Meets     Republican
 4 Abdnor, James             93  1973 SD        0      0      55    0.241         1  1923 0.110            0 Below     Republican
 5 Abercrombie, Neil        111  2009 HI        1      0      77   -0.432        10  1938 0.674            1 Below     Democrat
 6 Abercrombie, Neil        110  2007 HI        1      0      69   -0.432         9  1938 0.939            3 Meets     Democrat
 7 Abercrombie, Neil        109  2005 HI        1      0      63   -0.432         8  1938 0.311            0 Meets     Democrat
 8 Abercrombie, Neil        108  2003 HI        1      0      73   -0.432         7  1938 0.353            0 Meets     Democrat
 9 Abercrombie, Neil        107  2001 HI        1      0      69   -0.432         6  1938 0.390            0 Meets     Democrat
10 Abercrombie, Neil        106  1999 HI        1      0      62   -0.432         5  1938 0.427            1 Meets     Democrat
# ℹ 11,596 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Exercise

- Which member-Congress observations have the highest Legislative Effectiveness Score (`les`)?

Exercise

- Which member-Congress observations have the highest Legislative Effectiveness Score (`les`)?
- Which member-Congress observation passed the most bills through the House (`bills_passed`)?

Check your Answers

```r
arrange(cel, desc(les))
```

```
   member              congress  year state   dem female votepct ideology seniority  born   les bills_passed benchmark party
   <chr>                  <dbl> <dbl> <chr> <dbl>  <dbl>   <dbl>    <dbl>     <dbl> <dbl> <dbl>        <dbl> <fct>     <chr>
 1 Rangel, Charles          110  2007 NY        1      0      94   -0.514        19  1930  18.7           33 Exceeds   Democrat
 2 Murphy, John              96  1979 NY        1      0      54   -0.406         9  1926  17.9           31 Exceeds   Democrat
 3 Sullivan, Leonor          93  1973 MO        1      1      69   -0.386        11  1902  17.6           28 Exceeds   Democrat
 4 Sensenbrenner, F.        109  2005 WI        0      0      67    0.638        14  1943  17.6           25 Exceeds   Republican
 5 Murphy, John              95  1977 NY        1      0      NA   -0.406         8  1926  17.2           31 Exceeds   Democrat
 6 Ullman, Al                96  1979 OR        1      0      69   -0.367        12  1914  16.4           18 Exceeds   Democrat
 7 Smith, Lamar             112  2011 TX        0      0      69    0.425        13  1947  16.3           20 Exceeds   Republican
 8 Young, Don               108  2003 AK        0      0      75    0.284        16  1933  16.3           20 Exceeds   Republican
 9 Young, Don               107  2001 AK        0      0      70    0.284        15  1933  15.8           22 Exceeds   Republican
10 Young, Don               109  2005 AK        0      0      71    0.284        17  1933  15.3           18 Exceeds   Republican
# ℹ 11,596 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Check your Answers

```
arrange(cel, desc(bills_passed))
```

```
   member              congress  year state   dem female votepct ideology seniority  born   les bills_passed benchmark party
   <chr>                  <dbl> <dbl> <chr> <dbl>  <dbl>   <dbl>    <dbl>     <dbl> <dbl> <dbl>        <dbl> <fct>     <chr>
 1 Rangel, Charles          110  2007 NY        1      0      94  -0.514         19  1930  18.7           33 Exceeds   Democrat
 2 Murphy, John              95  1977 NY        1      0      NA  -0.406          8  1926  17.2           31 Exceeds   Democrat
 3 Murphy, John              96  1979 NY        1      0      54  -0.406          9  1926  17.9           31 Exceeds   Democrat
 4 Sullivan, Leonor          93  1973 MO        1      1      69  -0.386         11  1902  17.6           28 Exceeds   Democrat
 5 Sullivan, Leonor          94  1975 MO        1      1      75  -0.386         12  1902  12.7           25 Exceeds   Democrat
 6 Sensenbrenner, F.        109  2005 WI        0      0      67   0.638         14  1943  17.6           25 Exceeds   Republican
 7 Young, Don               107  2001 AK        0      0      70   0.284         15  1933  15.8           22 Exceeds   Republican
 8 Miller, George           110  2007 CA        1      0      84  -0.552         17  1945  10.6           22 Exceeds   Democrat
 9 Oberstar, James          110  2007 MN        1      0      64  -0.553         17  1934  13.0           22 Exceeds   Democrat
10 Rogers, Paul              93  1973 FL        1      0      60  -0.0850        10  1921  11.5           21 Exceeds   Democrat
# ℹ 11,596 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Choosing and Creating Columns

```r
select(cel, member, congress, les)
```

```
   member             congress    les
   <chr>                 <dbl>  <dbl>
 1 Abdnor, James            93 0.110
 2 Abzug, Bella             93 0.762
 3 Adams, Brock             93 1.24
 4 Addabbo, Joseph          93 0.155
 5 Albert, Carl             93 0.00103
 6 Alexander, Bill          93 1.88
 7 Anderson, John           93 0.289
 8 Anderson, Glenn          93 0.447
 9 Andrews, Ike             93 0.0879
10 Andrews, Mark            93 0.106
# ℹ 11,596 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Select Columns with select()

```r
select(cel, member:state)
```

```
   member             congress  year state
   <chr>                 <dbl> <dbl> <chr>
 1 Abdnor, James            93  1973 SD
 2 Abzug, Bella             93  1973 NY
 3 Adams, Brock             93  1973 WA
 4 Addabbo, Joseph          93  1973 NY
 5 Albert, Carl             93  1973 OK
 6 Alexander, Bill          93  1973 AR
 7 Anderson, John           93  1973 IL
 8 Anderson, Glenn          93  1973 CA
 9 Andrews, Ike             93  1973 NC
10 Andrews, Mark            93  1973 ND
# ℹ 11,596 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Select Columns with select()

```
select(cel, -(votepct:ideology))
```

```
   member            congress  year state   dem female seniority  born     les bills_passed benchmark party
   <chr>                <dbl> <dbl> <chr> <dbl>  <dbl>     <dbl> <dbl>   <dbl>        <dbl> <fct>     <chr>
 1 Abdnor, James           93  1973 SD        0      0         1  1923 0.110              0 Below     Republican
 2 Abzug, Bella            93  1973 NY        1      1         2  1920 0.762              1 Meets     Democrat
 3 Adams, Brock            93  1973 WA        1      0         5  1927 1.24               2 Meets     Democrat
 4 Addabbo, Joseph         93  1973 NY        1      0         7  1925 0.155              0 Below     Democrat
 5 Albert, Carl            93  1973 OK        1      0        14  1908 0.00103            0 Below     Democrat
 6 Alexander, Bill         93  1973 AR        1      0         3  1934 1.88               2 Meets     Democrat
 7 Anderson, John          93  1973 IL        0      0         7  1922 0.289              0 Meets     Republican
 8 Anderson, Glenn         93  1973 CA        1      0         3  1913 0.447              0 Meets     Democrat
 9 Andrews, Ike            93  1973 NC        1      0         1  1925 0.0879             0 Below     Democrat
10 Andrews, Mark           93  1973 ND        0      0         6  1926 0.106              0 Below     Republican
# ℹ 11,596 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Exercise

- Select only the `member`, `congress`, and `votepct` variables from `cel`

Exercise

- Select only the `member`, `congress`, and `votepct` variables from `cel`
- What does the `any_of()` function do? Why might it be useful with this vector?

```
vars <- c("member", "congress", "votepct")
```

Exercise

- Select only the `member`, `congress`, and `votepct` variables from `cel`
- What does the `any_of()` function do? Why might it be useful with this vector?

```
vars <- c("member", "congress", "votepct")
```

- What does this code produce?

```
select(cel, starts_with("b"))
```

Check your Answers

```
select(cel, member, congress, votepct)
```

```
   member             congress votepct
   <chr>                 <dbl>   <dbl>
 1 Abdnor, James            93      55
 2 Abzug, Bella             93      56
 3 Adams, Brock             93      85
 4 Addabbo, Joseph          93      75
 5 Albert, Carl             93      93
 6 Alexander, Bill          93     100
 7 Anderson, John           93      72
 8 Anderson, Glenn          93      75
 9 Andrews, Ike             93      50
10 Andrews, Mark            93      73
# ℹ 11,596 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Check your Answers

`any_of()` selects any of the named columns that exist, without erroring on ones that don't:

```r
vars <- c("member", "congress", "votepct")

select(cel, any_of(vars))
```

```
   member            congress votepct
   <chr>                <dbl>   <dbl>
 1 Abdnor, James           93      55
 2 Abzug, Bella            93      56
 3 Adams, Brock            93      85
 4 Addabbo, Joseph         93      75
 5 Albert, Carl            93      93
 6 Alexander, Bill         93     100
 7 Anderson, John          93      72
 8 Anderson, Glenn         93      75
 9 Andrews, Ike            93      50
10 Andrews, Mark           93      73
# ℹ 11,596 more rows
```

Check your Answers

All columns whose names start with "b": `born`, `bills_passed`, and `benchmark`

```
select(cel, starts_with("b"))
```

```
    born bills_passed benchmark
   <dbl>        <dbl> <fct>
 1  1923            0 Below
 2  1920            1 Meets
 3  1927            2 Meets
 4  1925            0 Below
 5  1908            0 Below
 6  1934            2 Meets
 7  1922            0 Meets
 8  1913            0 Meets
 9  1925            0 Below
10  1926            0 Below
# ℹ 11,596 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Add New Variables with mutate()

```r
mutate(cel, age = year - born)
```

```
   member            congress  year state   dem female votepct ideology seniority  born     les bills_passed benchmark party        age
   <chr>                <dbl> <dbl> <chr> <dbl>  <dbl>   <dbl>    <dbl>     <dbl> <dbl>   <dbl>        <dbl> <fct>     <chr>      <dbl>
 1 Abdnor, James           93  1973 SD        0      0      55   0.241          1  1923 0.110              0 Below     Republican    50
 2 Abzug, Bella            93  1973 NY        1      1      56  -0.597          2  1920 0.762              1 Meets     Democrat      53
 3 Adams, Brock            93  1973 WA        1      0      85  -0.396          5  1927 1.24               2 Meets     Democrat      46
 4 Addabbo, Joseph         93  1973 NY        1      0      75  -0.402          7  1925 0.155              0 Below     Democrat      48
 5 Albert, Carl            93  1973 OK        1      0      93  -0.392         14  1908 0.00103            0 Below     Democrat      65
 6 Alexander, Bill         93  1973 AR        1      0     100  -0.410          3  1934 1.88               2 Meets     Democrat      39
 7 Anderson, John          93  1973 IL        0      0      72   0.183          7  1922 0.289              0 Meets     Republican    51
 8 Anderson, Glenn         93  1973 CA        1      0      75  -0.269          3  1913 0.447              0 Meets     Democrat      60
 9 Andrews, Ike            93  1973 NC        1      0      50  -0.178          1  1925 0.0879             0 Below     Democrat      48
10 Andrews, Mark           93  1973 ND        0      0      73   0.0870         6  1926 0.106              0 Below     Republican    47
# ℹ 11,596 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Select, Manipulate, and Add New Variables with transmute()

```r
transmute(cel, member, congress, age = year - born)
```

```
   member             congress   age
   <chr>                 <dbl> <dbl>
 1 Abdnor, James            93    50
 2 Abzug, Bella             93    53
 3 Adams, Brock             93    46
 4 Addabbo, Joseph          93    48
 5 Albert, Carl             93    65
 6 Alexander, Bill          93    39
 7 Anderson, John           93    51
 8 Anderson, Glenn          93    60
 9 Andrews, Ike             93    48
10 Andrews, Mark            93    47
# ℹ 11,596 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Summaries and the Pipe

Create Summaries with summarise()

```
summarise(
  cel, avg_bills = mean(bills_passed), avg_seniority = mean(seniority)
)
```

	avg_bills	avg_seniority
	<dbl>	<dbl>
1	1.48	5.27

Creating Grouped Summaries with group_by() and summarise()

```
cel_party <- group_by(cel, party)

summarise(
  cel_party, avg_bills = mean(bills_passed), avg_seniority =
    mean(seniority)
)
```

	party	avg_bills	avg_seniority
	<chr>	<dbl>	<dbl>
1	Democrat	1.57	5.75
2	Republican	1.38	4.72

Exercise

- Calculate each member's age in each Congress

Exercise

- Calculate each member's age in each Congress
- Find each member's average number of bills passed across all Congresses in the `cel` dataset

Exercise

- Calculate each member's age in each Congress
- Find each member's average number of bills passed across all Congresses in the `cel` dataset
- Find the member with the lowest average bills passed across these Congresses

Exercise

- Calculate each member's age in each Congress
- Find each member's average number of bills passed across all Congresses in the `cel` dataset
- Find the member with the lowest average bills passed across these Congresses
- Find the member with the highest average bills passed across these Congresses

Check your Answers

```r
mutate(cel, age = year - born)
```

```
   member           congress  year state   dem female votepct ideology seniority  born     les bills_passed benchmark party        age
   <chr>               <dbl> <dbl> <chr> <dbl>  <dbl>   <dbl>    <dbl>     <dbl> <dbl>   <dbl>        <dbl> <fct>     <chr>      <dbl>
 1 Abdnor, James          93  1973 SD        0      0      55   0.241          1  1923 0.110              0 Below     Republican    50
 2 Abzug, Bella           93  1973 NY        1      1      56  -0.597          2  1920 0.762              1 Meets     Democrat      53
 3 Adams, Brock           93  1973 WA        1      0      85  -0.396          5  1927 1.24               2 Meets     Democrat      46
 4 Addabbo, Joseph        93  1973 NY        1      0      75  -0.402          7  1925 0.155              0 Below     Democrat      48
 5 Albert, Carl           93  1973 OK        1      0      93  -0.392         14  1908 0.00103            0 Below     Democrat      65
 6 Alexander, Bill        93  1973 AR        1      0     100  -0.410          3  1934 1.88               2 Meets     Democrat      39
 7 Anderson, John         93  1973 IL        0      0      72   0.183          7  1922 0.289              0 Meets     Republican    51
 8 Anderson, Glenn        93  1973 CA        1      0      75  -0.269          3  1913 0.447              0 Meets     Democrat      60
 9 Andrews, Ike           93  1973 NC        1      0      50  -0.178          1  1925 0.0879             0 Below     Democrat      48
10 Andrews, Mark          93  1973 ND        0      0      73   0.0870         6  1926 0.106              0 Below     Republican    47
# ℹ 11,596 more rows
# ℹ Use `print(n = ...)` to see more rows
```

```
cel_member <- group_by(cel, member)

avg_bills_by_member <- summarise(cel_member, avg_bills =
    mean(bills_passed))

avg_bills_by_member
```

```
   member                 avg_bills
   <chr>                      <dbl>
 1 Abdnor, James              0.5
 2 Abercrombie, Neil          0.545
 3 Abraham, Ralph             1.33
 4 Abzug, Bella               1.5
 5 Acevedo-Vila, Anibal       2
 6 Ackerman, Gary             1.33
 7 Adams, Alma                1.33
 8 Adams, Brock               1.5
 9 Adams, Sandy               2
10 Addabbo, Joseph            1.57
# ℹ 2,223 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Check your Answers

```
arrange(avg_bills_by_member, avg_bills)
```

```
   member              avg_bills
   <chr>                   <dbl>
 1 Albert, Carl                0
 2 Ammerman, Joseph            0
 3 Amo, Gabe                   0
 4 Arends, Leslie              0
 5 Ashbrook, John              0
 6 Austria, Steve              0
 7 Bacchus, Jim                0
 8 Badillo, Herman             0
 9 Baesler, Scotty             0
10 Bafalis, L.                 0
# ℹ 2,223 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Check your Answers

```r
arrange(avg_bills_by_member, desc(avg_bills))
```

```
   member            avg_bills
   <chr>                 <dbl>
 1 Sullivan, Leonor       26.5
 2 Rogers, Paul           19.7
 3 Murphy, John           17
 4 Ullman, Al             14.8
 5 Rodino, Peter          12.1
 6 Mahon, George          12
 7 Hays, Wayne            11
 8 Katko, John            10.5
 9 Brooks, Jack           10.2
10 Jones, Robert          10
# ℹ 2,223 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Utilize Multiple Operations with the Pipe

- We created a lot of different objects during our calculations

Utilize Multiple Operations with the Pipe

- We created a lot of different objects during our calculations
- Can we avoid creating unneeded objects?

Utilize Multiple Operations with the Pipe

- We created a lot of different objects during our calculations
- Can we avoid creating unneeded objects?
- We can utilize a feature of the tidyverse: the pipe

Utilize Multiple Operations with the Pipe

- Pipe can be read as:

Utilize Multiple Operations with the Pipe

- Pipe can be read as:
- Take this `|>` (and then...)

Utilize Multiple Operations with the Pipe

- Pipe can be read as:
- Take this `|>` (and then...)
- do this `|>` (and then...)

Utilize Multiple Operations with the Pipe

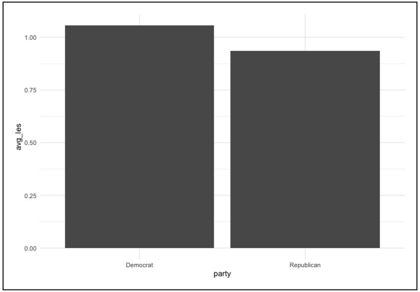
- Pipe can be read as:
- Take this |> (and then...)
- do this |> (and then...)
- do this

```r
cel |>
  group_by(state) |>
  summarise(avg_les = mean(les)) |>
  arrange(desc(avg_les))
```

```
   state avg_les
   <chr>   <dbl>
 1 AK       4.87
 2 DC       2.65
 3 OR       1.54
 4 MT       1.52
 5 UT       1.50
 6 NM       1.49
 7 NJ       1.40
 8 AZ       1.39
 9 DE       1.30
10 MS       1.28
# ℹ 46 more rows
# ℹ Use `print(n = ...)` to see more rows
```

Utilize Multiple Operations with the Pipe

```r
cel |>
  group_by(party) |>
  summarise(avg_les = mean(les)) |>
  ggplot(aes(x = party, y = avg_les)) +
  geom_col() +
  theme_minimal()
```



A Note on the Pipe

- Base pipe:

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- `|>`
- Can be used without loading any packages

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- Base pipe:
- `|>`
- Can be used without loading any packages
- Relatively new: introduced in 2021

A Note on the Pipe

- Tidyverse pipe:

A Note on the Pipe

- Tidyverse pipe:
- %>%

A Note on the Pipe

- Tidyverse pipe:
- `%>%`
- Must load `dplyr` or `magrittr` to use

Exercise

- Calculate the average Legislative Effectiveness Score for Maryland members serving in Congresses after the 110th (2007 onward). Display only the member and average LES variables

Exercise

- Calculate the average Legislative Effectiveness Score for Maryland members serving in Congresses after the 110th (2007 onward). Display only the member and average LES variables
- Plot the results

Check your Answers

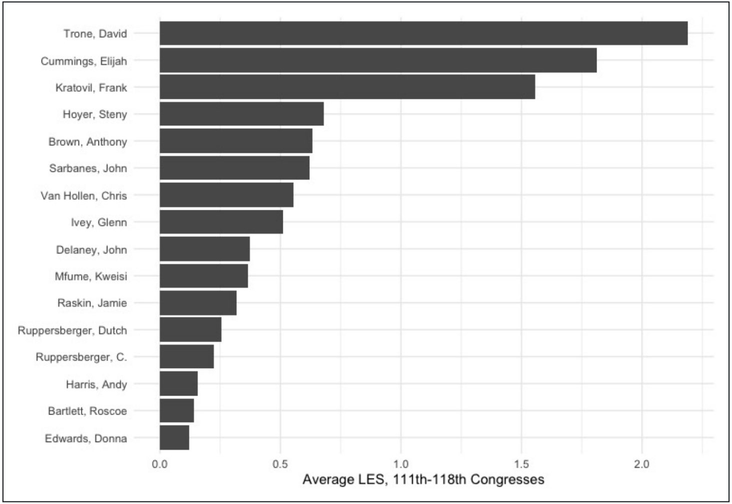
```r
cel |>
  filter(state == "MD", congress > 110) |>
  group_by(member) |>
  summarise(avg_les = mean(les))
```

	member	avg_les
	<chr>	<dbl>
1	Bartlett, Roscoe	0.139
2	Brown, Anthony	0.634
3	Cummings, Elijah	1.81
4	Delaney, John	0.374
5	Edwards, Donna	0.122
6	Harris, Andy	0.156
7	Hoyer, Steny	0.679
8	Ivey, Glenn	0.509
9	Kratovil, Frank	1.56
10	Mfume, Kweisi	0.366
11	Raskin, Jamie	0.317
12	Ruppersberger, C.	0.222
13	Ruppersberger, Dutch	0.254
14	Sarbanes, John	0.622
15	Trone, David	2.19
16	Van Hollen, Chris	0.552

Check your Answers

```
cel |>
  filter(state == "MD", congress > 110) |>
  group_by(member) |>
  summarise(avg_les = mean(les)) |>
  ggplot(aes(x = avg_les, y = fct_reorder(member, avg_les))) +
  geom_col() +
  theme_minimal() +
  labs(x = "Average LES, 111th-118th Congresses", y = NULL)
```

Check your Answers



Summary

- Familiarized yourself with basic `R` syntax

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- Learned how to manipulate your data

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- Familiarized yourself with basic `R` syntax
- Learned how to manipulate your data
- Wrote concise code that is easy to follow

- On Thursday, you will apply the skills learned in this course to an idea that interests you. In doing so, you will need to find a dataset that:

Homework

- On Thursday, you will apply the skills learned in this course to an idea that interests you. In doing so, you will need to find a dataset that:
- Is relevant to your research interests

Homework

- On Thursday, you will apply the skills learned in this course to an idea that interests you. In doing so, you will need to find a dataset that:
- Is relevant to your research interests
- Contains continuous and categorical variables

- On Thursday, you will apply the skills learned in this course to an idea that interests you. In doing so, you will need to find a dataset that:
- Is relevant to your research interests
- Contains continuous and categorical variables
- If needed, reach out to me for dataset recommendations

Questions?